

What's special about 89% credibility intervals

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A curious property of 89% credibility intervals is discussed.

A Bayesian *credibility interval*, or credible interval, is a range where the true value of some real quantity lies, with a given probability. For instance, if $[1.2, 7.0]$ is a 60% credibility interval for the quantity Q , then there is a 60% probability that the true value of Q is between 1.2 and 7.4 (note the difference in meaning from a *confidence* interval, whose definition is much more involved and less direct).

It is of course always best to report the full posterior probability distribution of a quantity of interest, so that the reader can derive from it the probabilities of any desired range as well as other properties. But a credibility interval often provides a useful summary of the full probability distribution.

Credibility intervals of 50%, 75%, 90%, 95%, or even 99% probability are more or less commonly reported in the Bayesian literature. Since at least 2016 also 89% credibility intervals have become quite fashionable. They seem to originate in McElreath's *Statistical Rethinking* ([mcelreath2016_r2020](#)), first published in that year. McElreath rightly points out that there is nothing

The result of a Bayesian inference about some quantity of interest is usually a full probability distribution over the possible values of that quantity. From such a distribution the reader can then, for instance, make decisions by combining the distribution with appropriate utilities; this includes the decision on which value should be reported, if a unique answer ('estimate') is required.

But it is often useful to pro

1

¹ [mcelreath2016_r2020](#).